

Git, GitHub, and Codespaces

A hands-on tutorial: build a Python project from scratch

From an empty repository to a tested, synchronized project

No installations. No third-party Python packages.

What we will build—and what you will be able to do

Our finished repository

```
growth-calculator/  
  README.md  
  growth.py  
  test_growth.py  
  .gitignore
```

By the end, you can

- ▶ create a repository and a Codespace;
- ▶ edit, run, and test Python code;
- ▶ inspect, stage, and commit changes;
- ▶ push work to GitHub and pull new work;
- ▶ recognize the state of a Git repository.

What this means: We will learn Git by repeatedly using it on one small project.

Why not save a new file every time?

```
growth.py  
growth_final.py  
growth_final_v2.py  
growth_final_v2_fixed.py  
growth_REALLY_FINAL.py
```

This does not tell us:

- ▶ what changed between versions;
- ▶ why a change was made;
- ▶ which code generated a result;
- ▶ whether two collaborators changed the same file.

What this means: Git stores named snapshots of the project and a history connecting them.

Three tools, three different jobs

Git

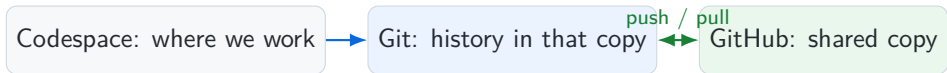
Records versions of files and connects them in a history.

GitHub

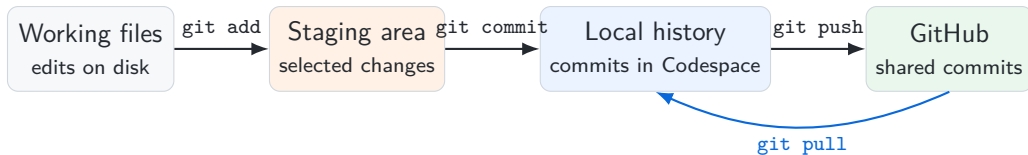
Hosts a repository online so it can be shared and synchronized.

Codespaces

Provides a cloud computer and VS Code editor in the browser.



The Git pipeline: where is a change?



The distinction that prevents most confusion

Saving a file, committing it, and pushing it are three separate actions.

Step 1: open GitHub's repository form

In the GitHub website

1. Sign in at `github.com`.
2. In the upper-right corner, click .
3. Select .



What this means: A repository is the project folder plus the Git history attached to it.

Step 2: configure and create the repository

Create a new repository

Owner	your-username
Repository name	growth-calculator
Description	A simple Python growth calculator
Visibility	<i>circ</i> Public ● Private
Add a README file	☒
Create repository	

Checkpoint: the repository exists on GitHub

Why add a README?

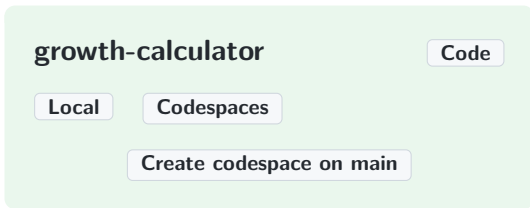
- ▶ It explains the project to a visitor.
- ▶ It gives the repository a first file.
- ▶ GitHub creates the first commit for us.

Step 3: start a Codespace

From the repository page

1. Click the green `Code` button.
2. Select the `Codespaces` tab.
3. Click `Create codespace on main`.

GitHub opens a new browser tab.
Starting the cloud computer can take a minute.



What this means: The Codespace contains a working copy of the GitHub repository.

Orient yourself inside Codespaces



The areas we will use

- ▶ **Explorer:** create and open files.
- ▶ **Editor:** write code.
- ▶ **Terminal:** run commands.

Open the terminal with:

Terminal → New Terminal

Inspect the starting repository

Type in the terminal

```
$ pwd
/workspaces/growth-calculator
$ ls
README.md
$ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
```

How to read this

- ▶ `pwd`: show the current folder.
- ▶ `ls`: list the files in it.
- ▶ `main`: the current branch.
- ▶ `origin/main`: GitHub's corresponding branch.
- ▶ `clean`: no unsaved Git changes.

Checkpoint: both copies are synchronized

Read the existing history

Type

```
$ git log --oneline  
a1b2c3d (HEAD -> main, origin/main)  
Initial commit
```

Interpret the output

- ▶ a1b2c3d: shortened commit identifier.
- ▶ HEAD -> main: where our Codespace currently points.
- ▶ origin/main: where GitHub currently points.
- ▶ Initial commit: the saved description.

What this means: Both labels point to the same commit, so the two copies are synchronized.

Create the program: growth.py

In the Explorer

1. Click the **New File** icon.
2. Name the file growth.py.
3. Enter the code shown here.
4. Save with **Ctrl+S**.

No package installation is needed.

```
def percentage_change(old_value, new_value):  
    """Return percentage change from old to new."""  
    if old_value == 0:  
        raise ValueError("old_value cannot be zero")  
  
    return 100 * (new_value - old_value) / old_value  
  
if __name__ == "__main__":  
    result = percentage_change(100, 105)  
    print(f"Percentage change: {result}%")
```

Run the program before saving a Git version

Type

```
$ python growth.py  
Percentage change: 5.0%
```

Try a different pair of values by changing:

```
result = percentage_change(100, 105)
```

What just happened?

- ▶ Python executed the file.
- ▶ The program produced the expected result.
- ▶ Git has not saved a new version.
- ▶ GitHub has not received anything.

What this means: Running code and recording its version are separate actions.

Ask Git what changed

```
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what
   will be committed)
    growth.py

nothing added to commit
```

“Untracked” means

- ▶ the file exists in the Codespace;
- ▶ Git sees it;
- ▶ it has never been included in a commit.

A useful subtlety

`git diff` shows changes to tracked files. It will not display the contents of this new, untracked file.

Stage the file, then review exactly what is staged

```
$ git add growth.py
$ git status
Changes to be committed:
  new file:   growth.py

$ git diff --staged
diff --git a/growth.py b/growth.py
new file mode 100644
--- /dev/null
+++ b/growth.py
@@ ...
+def percentage_change(old_value, new_value):
```

What changed in Git?

- ▶ `git add` selected `growth.py` for the next commit.
- ▶ It did not create the commit.
- ▶ `git diff --staged` previews the exact snapshot.

State: staged, not committed

Commit: save one coherent change

```
$ git commit -m "Add growth calculator"
[main d4e5f6a] Add growth calculator
1 file changed, 11 insertions(+)
create mode 100644 growth.py

$ git log --oneline
d4e5f6a (HEAD -> main) Add growth calculator
a1b2c3d (origin/main) Initial commit
```

How to read this

- ▶ The commit identifier is d4e5f6a.
- ▶ HEAD -> main moved to the new commit.
- ▶ origin/main did not move.
- ▶ The commit exists locally, not yet on GitHub.

State: committed locally

Push: send local commits to GitHub

```
$ git push
Enumerating objects: 4, done.
Writing objects: 100% (3/3), done.
To https://github.com/your-username/
growth-calculator.git
   a1b2c3d..d4e5f6a  main -> main
```

Then return to the GitHub tab and refresh the repository page.

Checkpoint: Codespace and GitHub contain the new commit

Verify on GitHub

1. Is `growth.py` visible?
2. Is the commit message visible?
3. Does the commit history show two commits?

Why verification matters

A successful local commit does not imply a successful push.

Create package-free tests: test_growth.py

Create a new file named `test_growth.py`. Each `assert` states a result that must be true. If one is false, Python stops with an error.

No installed packages

`growth` refers to our own `growth.py` file. The tests use only Python itself.

```
from growth import percentage_change

assert percentage_change(100, 105) == 5
assert percentage_change(100, 90) == -10
assert percentage_change(100, 100) == 0

print("All tests passed.")
```

Run the tests—and see what failure looks like

When all expectations are correct

```
$ python test_growth.py  
All tests passed.
```

Temporarily change the first expected result from 5 to 6 and rerun.

A failing test

```
$ python test_growth.py  
Traceback (most recent call last):  
  File "test_growth.py", line 4  
    assert percentage_change(100,105) == 6  
AssertionError
```

Restore 5 and rerun until all tests pass.

What this means: A test converts an expected result into a check the computer can repeat.

Record the tested change

```
$ git status
$ git diff
$ python test_growth.py
All tests passed.
$ git add test_growth.py
$ git diff --staged
$ git commit -m "Add tests for growth calculator"
$ git push
```

Why this order?

1. Identify changed files.
2. Inspect their contents.
3. Verify the code works.
4. Select the intended change.
5. Review, commit, and share it.

What this means: A good commit is small, coherent, reviewed, and tested.

Simulate a collaborator by editing on GitHub

On the GitHub repository page

1. Click README.md.
2. Click the pencil icon `Edit this file`.
3. Add the text shown at right.
4. Click `Commit changes...`.
5. Enter Describe the project and confirm.

Now GitHub is one commit ahead of the Codespace

README.md

Edit

```
# Growth calculator  
A small Python program that  
calculates percentage changes.
```

Fetch, inspect, then pull the new commit

```
$ git fetch
From https://github.com/.../growth-calculator
6141a87..5f89e4d  main -> origin/main

$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit,
and can be fast-forwarded.
  (use "git pull" to update your local branch)

$ git pull
Updating 6141a87..5f89e4d
Fast-forward
 README.md | 3 +++
 1 file changed, 3 insertions(+)
```

What each command does

1. `git fetch` contacts GitHub and refreshes `origin/main`; it does not change your files.
2. `git status` now sees that GitHub is one commit ahead.
3. `git pull` moves `main` forward and updates `README.md`.

“Fast-forward” means there was no competing local commit to merge.

What is .gitignore?

A plain-text instruction file

We create .gitignore to tell Git which files or folders it should not track.

Each line names a path or pattern to ignore. That line is sometimes called a “rule.”

Nothing is deleted

The directory stays on the computer. Git simply stops offering to include it in commits.

For this project, the complete file is:

```
__pycache__/
```

This one line means: “Git, ignore the __pycache__/ directory that Python created.”

Create, check, and commit .gitignore

1. Create the file

```
$ printf "__pycache__/\n" > .gitignore  
$ cat .gitignore  
__pycache__/
```

printf writes the line into .gitignore;
cat checks what was written.

2. Check the result

```
$ git status --short  
?? .gitignore
```

The cache is no longer listed because Git is ignoring it.

3. Share the instruction

```
$ git add .gitignore  
$ git commit -m "Ignore Python cache"  
$ git push
```

We commit .gitignore so everyone using the repository gets the same instruction.

Tracked: .gitignore
Ignored: __pycache__/

If a push is rejected, synchronize first

```
$ git push
! [rejected] main -> main (fetch first)
error: failed to push some refs
```

This means GitHub has commits that your Codespace does not have. Git refuses to overwrite them.

Safe response

```
$ git status
$ git pull
# Read any message from Git carefully.
$ python test_growth.py
$ git push
```

If Git reports a conflict, stop and resolve the marked files before pushing.

Do not solve this with force

For this course workflow, do not use `git push -force`: it can discard shared history.

What is a merge conflict?

A conflict occurs when Git cannot decide how to combine competing edits automatically.

Git marks the disputed region:

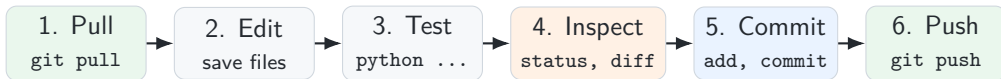
```
«««< HEAD
Percentage change calculator
=====
Growth rate calculator
»»»> origin/main
```

Resolution procedure

1. Read both versions.
2. Edit the file to the intended final content.
3. Remove all conflict markers.
4. Run the program and tests.
5. Stage and commit the resolution.

What this means: A conflict is a request for human judgment, not evidence that work was lost.

The daily workflow



Before the commit

- ▶ Does the program run?
- ▶ Do the tests pass?
- ▶ Did I inspect every changed line?

After the commit

- ▶ Is the message meaningful?
- ▶ Did the push succeed?
- ▶ Can I see the commit on GitHub?

Your turn: make one complete change

Choose one improvement

- ▶ Add a test using values 200 and 210.
- ▶ Change the displayed example.
- ▶ Improve the function's docstring.
- ▶ Add instructions to README.md.

Done means: the new commit is visible on GitHub

Complete the whole cycle

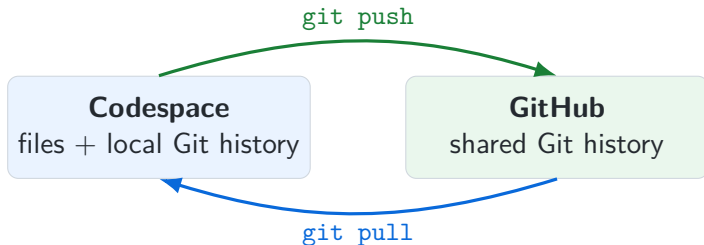
```
$ git pull
# Edit and save the chosen file.
$ python growth.py
$ python test_growth.py
$ git status
$ git diff
$ git add <the-file-you-changed>
$ git diff --staged
$ git commit -m "Describe the improvement"
$ git push
```

Command guide: question first, command second

Question	Command
What is the repository's current state?	<code>git status</code>
What unstaged tracked lines changed?	<code>git diff</code>
What exactly is selected for the next commit?	<code>git diff -staged</code>
Which changes should enter the next commit?	<code>git add <file></code>
How do I save the staged snapshot?	<code>git commit -m "message"</code>
What commits exist in the history?	<code>git log -oneline</code>
How do I receive GitHub's newer commits?	<code>git pull</code>
How do I send my local commits to GitHub?	<code>git push</code>

What this means: When uncertain, start with `git status`; read Git's message before acting.

The one picture to remember



Pull → Edit → Test → Inspect → Commit → Push

Use it on every project until the workflow becomes routine.